

PAPER_1326 — Supermassive BH Seed Mass via Direct Collapse

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "smbh_seeds"})`

Master Expression

$$M_{\text{seed}} = A_5 \times D_{\text{BSFG}}^2 \times D_{\text{crit}} = 56,160 M_{\text{sun}}$$

Match: within obs 10^4 - $10^5 M_{\text{sun}}$

Interpretation

Direct collapse seeds via DPM buoyancy at $z > 15$. Explains $z \sim 10$ SMBH observations.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 11, 2026, Youngstown, OH.